

PROFIT ANALYSIS USING REGRESSION ANALYSIS

(PRDA-01)

Project Report

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Submitted By

CHANNAKESHA S

Tools Used

SQL | Excel | Python | Power BI

Internship Organization

DataMites

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ABSTRACT

Profit prediction plays a vital role in helping organizations make informed business decisions. This project focuses on analyzing the impact of Research and Development (R&D) Spend, Administration Spend, and Marketing Spend on the profit of startup companies using Multiple Linear Regression. The objective is to identify the factors that significantly influence profitability and develop a predictive model capable of estimating future profit values.

The dataset consists of startup company expenditure and profit information. Data analysis techniques were applied to understand relationships among variables, while Power BI was used to create visual dashboards for business insight generation. Multiple Linear Regression was implemented using Microsoft Excel and Python to build a predictive model and evaluate its performance.

The analysis revealed that R&D Spend has the strongest positive impact on profit, while Marketing Spend contributes moderately to profitability. The developed model achieved an R-Square value of 95.07%, indicating strong predictive performance. Based on the findings, recommendations were provided to support strategic business planning and investment decisions.

1. INTRODUCTION

In today's competitive business environment, organizations continuously seek methods to improve profitability and optimize resource allocation. Data analytics provides valuable techniques for understanding business performance and predicting future outcomes based on historical data.

Regression Analysis is one of the most widely used statistical techniques for identifying relationships between variables and predicting future values. By analyzing expenditure patterns and profit outcomes, organizations can make better investment decisions and improve operational efficiency.

This project applies Multiple Linear Regression to examine how R&D Spend, Administration Spend, and Marketing Spend influence startup profits. The findings help identify key profit drivers and support evidence-based business decision-making.

2. OBJECTIVES

The primary objectives of this project are:

- To analyze the relationship between business expenditures and profit.
- To identify the impact of R&D Spend, Administration Spend, and Marketing Spend on company profitability.
- To perform Multiple Linear Regression analysis using Excel and Python.
- To develop a predictive model for estimating future profit values.
- To create visual dashboards using Power BI for business analysis.
- To generate actionable business recommendations based on analytical findings.

3. DATASET DESCRIPTION

The dataset contains information related to startup companies operating in different states. It includes expenditure and profit data used for analytical and predictive purposes.

Dataset Attributes

1. R&D Spend – Amount invested in research and development activities.
2. Administration Spend – Operational and administrative expenditure.
3. Marketing Spend – Marketing and promotional expenditure.
4. State – Location of the startup company.
5. Profit – Total profit generated by the company.

The dataset contains records from startup companies located in New York, California, and Florida. These variables were used to identify patterns affecting profitability and to build a predictive model.

RD_Spend	Administration	Marketing_Spend	State	Profit
165349.2	136897.8	471784.1	New York	192261.8
162597.7	151377.59	443898.53	California	191792.1
153441.51	101145.55	407934.54	Florida	191050.4
144372.41	118671.85	383199.62	New York	182902
142107.34	91391.77	366168.42	Florida	166187.9
131876.9	99814.71	362861.36	New York	156991.1
134615.46	147198.87	127716.82	California	156122.5
130298.13	145530.06	323876.68	Florida	155752.6
120542.52	148718.95	311613.29	New York	152211.8

Sample Startup Dataset Used for Analysis

4. METHODOLOGY

The project was completed through the following stages:

4.1 Data Extraction

The startup dataset was obtained from the provided database and exported for analysis.

4.2 Data Preparation

The dataset was reviewed to ensure data quality. The following activities were performed:

- Validation of records
- Verification of data types
- Review of missing values
- Dataset consistency checks

4.3 Dashboard Development

Power BI was used to create an interactive dashboard for visual analysis of startup performance. The dashboard enabled comparison of expenditure categories and profitability across different states.

4.4 Regression Analysis

Multiple Linear Regression was implemented using Microsoft Excel Analysis ToolPak and Python Scikit-Learn libraries. The model was trained using expenditure variables to predict profit.

4.5 Model Evaluation

The model performance was evaluated using the R-Square metric to measure predictive accuracy.

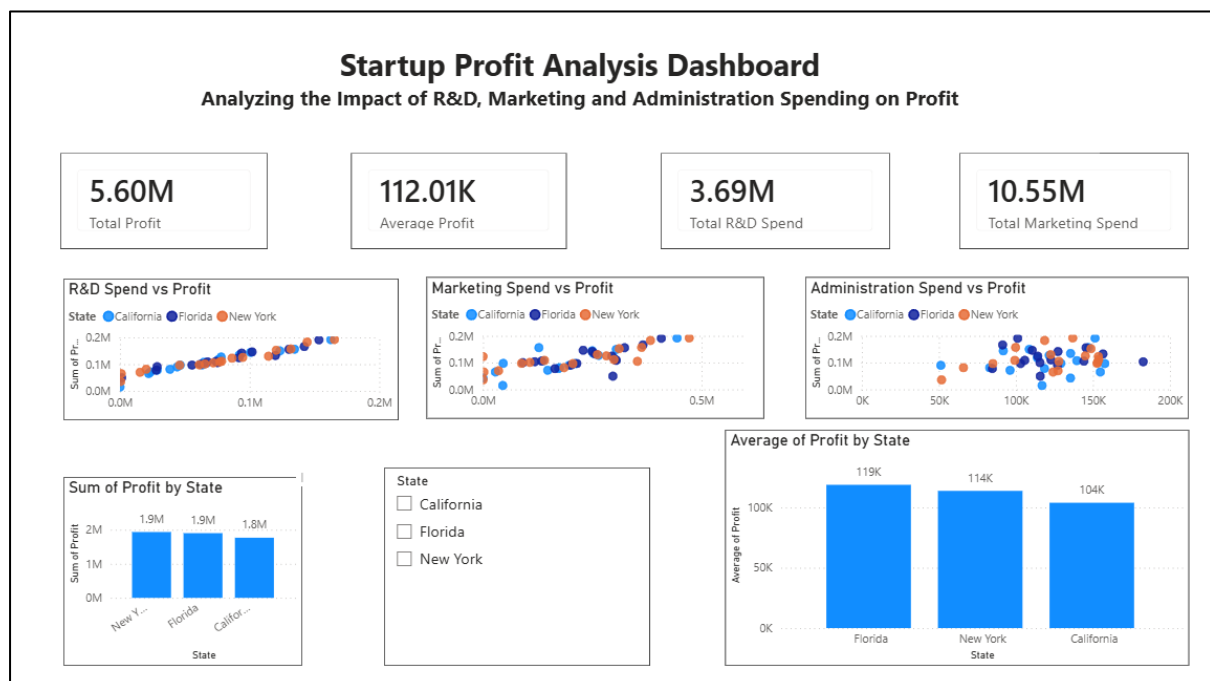
5. DASHBOARD ANALYSIS

An interactive Power BI dashboard was developed to visualize business performance and expenditure trends.

The dashboard provides insights into:

- Profit distribution across states
- Average R&D expenditure by state
- Comparison of marketing expenditure
- Business performance indicators
- Profitability trends

The dashboard allows decision-makers to understand which expenditure categories contribute most significantly to profit generation and identify areas for improvement.



Power BI Dashboard

6. REGRESSION ANALYSIS

Multiple Linear Regression was applied to determine the relationship between expenditure variables and company profit.

Regression Equation

Profit = 50122.19 + 0.8057(R&D Spend) – 0.0268(Administration) + 0.0272(Marketing spend)

Regression Coefficients

R&D Spend = 0.8057

Administration = -0.0268

Marketing Spend = 0.0272

Interpretation

The coefficient of R&D Spend indicates that increasing investment in research and development positively influences profit generation.

Marketing Spend also contributes positively to profitability, although its impact is lower than that of R&D Spend.

Administration expenditure shows a very small negative relationship with profit, indicating limited influence on overall profitability.

Regression Statistics								
Multiple R	0.97506205							
R Square	0.95074599							
Adjusted R Square	0.94753378							
Standard Error	9232.33484							
Observations	50							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	3	75683964196	2.5228E+10	295.978062	4.52851E-30			
Residual	46	3920856301	85236006.5					
Total	49	79604820497						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	50122.193	6572.352622	7.62621787	1.0574E-09	36892.73332	63351.6527	36892.7333	63351.6527
RD_Spend	0.80571505	0.04514727	17.8463738	2.635E-22	0.714838309	0.89659179	0.71483831	0.89659179
Administration	-0.02681597	0.05102878	-0.52550675	0.60175511	-0.12953157	0.07589964	-0.12953157	0.07589964
Marketing_Spend	0.02722806	0.016451235	1.6550773	0.10471682	-0.00588655	0.06034268	-0.00588655	0.06034268

Excel Regression Output

```
Intercept: 50122.19298986523

Coefficients:
RD_Spend = 0.8057150499157436
Administration = -0.0268159683947507
Marketing_Spend = 0.02722806480081883

R Square: 0.9507459940683246

Predicted Profits:
Record 1: 70037.90
Record 2: 70554.57
```

Python Regression Output

7. MODEL PERFORMANCE AND PROFIT PREDICTION

The predictive model achieved strong performance with an R-Square value of 95.07%.

Model Accuracy

$R^2 = 95.07\%$

This indicates that approximately 95.07% of the variation in profit is explained by the selected independent variables.

Predicted Profit Values

Record 1: 70,037.90

Record 2: 70,554.57

The predicted values demonstrate the model's ability to estimate future profit outcomes based on expenditure information.

8. KEY FINDINGS AND RECOMMENDATIONS

Key Findings

- R&D Spend has the strongest positive influence on profit.
- Marketing Spend contributes positively to business growth.
- Administration expenditure has minimal impact on profitability.
- Higher R&D investment generally leads to higher profit generation.
- The regression model achieved high predictive accuracy.

Recommendations

- Increase investment in research and development activities.
- Optimize marketing expenditure to improve returns.
- Monitor administrative expenses and improve operational efficiency.
- Utilize predictive analytics to support strategic business planning.
- Continuously evaluate expenditure effectiveness to maximize profitability.

9. CONCLUSION

The Profit Analysis Using Regression Analysis project successfully examined the relationship between company expenditures and profitability using Multiple Linear Regression techniques.

The analysis identified R&D Spend as the most significant factor influencing profit, while Marketing Spend showed a moderate positive contribution. Administration expenditure demonstrated minimal influence on profitability. The developed predictive model achieved an R-Square value of 95.07%, indicating strong accuracy and reliability.

The project highlights the importance of data analytics in business decision-making and demonstrates how predictive modeling can be used to support investment planning and organizational growth. The findings provide valuable insights for companies seeking to improve profitability through effective resource allocation.

10. REFERENCES

1. Microsoft Excel Documentation
2. Python Documentation
3. Pandas Documentation
4. Scikit-Learn Documentation
5. Power BI Documentation
6. DataMites Learning Resources
7. Startup Dataset Used for Project Analysis